Machine learning is a subset of artificial intelligence that uses data and algorithms to let computers learn and make predictions without being explicitly programmed. [1, 2]

How Machine Learning Works

Machine learning works by feeding large amounts of data into an algorithm to find hidden patterns. The process has two main steps: [1, 2]

- **Training:** The system looks at examples to learn rules and adjust its internal settings.
- **Inference:** The trained model uses what it learned to make accurate predictions on new, unseen data. [1]

Main Types of Machine Learning

- **Supervised Learning:** The model trains on labeled data, where both the input and correct output are provided (e.g., spam filters).
- **Unsupervised Learning:** The model analyzes unlabeled data to find hidden structures or groupings on its own (e.g., customer clustering).
- **Reinforcement Learning:** An agent learns to make decisions through trial and error, using rewards and penalties (e.g., game-playing AI).
- **Semi-Supervised Learning:** A mix of a small amount of labeled data and a large pool of unlabeled data is used to improve accuracy. [1, 2]

Real-World Applications

- **Recommendation Systems:** Platforms like Netflix and Spotify suggest movies or music based on your past activity.
- **Natural Language Processing (NLP):** Chatbots and virtual assistants understand and generate human speech.
- **Computer Vision:** Self-driving cars and medical imaging tools interpret visual inputs to spot objects or abnormalities.
- **Fraud Detection:** Banks spot strange transactions to stop credit card theft before it harms users. [1, 2]